Library Management System

Title description

In a magical library, there are various books, each with its unique information. Please design a *Book* class and a *Library* class, with the following requirements:
Both *Book* and *Library* need to use class templates, so that objects with integer attributes and objects with real attributes can be created.
The *Book* class is initialized by passing in the book name (string), author (string), and page number (integer or real).
The *Library* class can add books and manage them.
The *Book* class has the following two interfaces:
void show(): Outputs the book's name, author, and page number in the format: "Book name: title, Author: author, Pages: pages", and ends with a newline character.
bool operator <(const Book<T>& other): Overloading the < operator to compare the page numbers of two books.
The *Library* class has the following three interfaces:
void addBook(const Book<T>& book): Adds a book to the library.
void show(): Outputs the information of all books in the library, with each book's information occupying a separate line. The output order must be consistent with the order of addition.
void borrowBook(const std::string& title): Pass in the book name, judge whether the book exists in the library, and output the corresponding results. If so, output "Books Checked out title."; otherwise output "Book title does not exist.". Note that this function only checks whether the book exists, and does not need to remove the book from the library. The total number of books is not large, and sequential search can also be AC.
In this problem, we assume that the data type of the page number is the same.
An example of using the *Book* class and *Library* class is as follows:

```
#include<iostream>
#include"Book.h"
#include"Library.h"
using namespace std;

int main() {

    Library<int> library;
    Book<int> book1("The Catcher in the Rye", "J.D. Salinger", 277);
    Book<int> book2("To Kill a Mockingbird", "Harper Lee", 324);
```

```
    library.addBook(book1);
    library.addBook(book2);
    library.show();
    // The output should be:
    // Book name: The Catcher in the Rye, Author: J.D. Salinger, Pages: 277
    // Book name: To Kill a Mockingbird, Author: Harper Lee, Pages: 324

    library.borrowBook("The Catcher in the Rye");
    // The output should be: Books Checked out The Catcher in the Rye.

    library.borrowBook("1984");
    // The output should be: Book 1984 does not exist.

    Book<int> book3("1984", "George Orwell", 328);
    if (book2 < book3) book2.show();
    else book3.show();
    // The output should be: Book name: To Kill a Mockingbird, Author: Harper
Lee, Pages: 324

    return 0;}
```

Submission format

You need to submit multiple files, including *Book.h, Library.h* files; main.cpp and Makefile files are not required to be uploaded.
You should compress your files into a zip file and upload it. Note: your files should be in the root directory of the compressed package, not in a subfolder of the compressed package. During evaluation, OJ will compile and execute your code by pasting the provided main.cpp into your directory.